Machine Learning (2026) 115:240
https://doi.org/10.1007/s10994-026-07181-0

Beyond Lipschitz: generalization and optimization analysis for adversarial training under Hölder-type conditions

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Received: 14 March 2026 / Revised: 25 August 2026 / Accepted: 25 September 2026
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Abstract
Adversarial training improves model robustness but may exhibit robust overfitting. Previous theoretical analyses mostly emphasize Lipschitz continuity of the adversarial loss, while understanding under weaker smoothness conditions (often observed in practice) remains limited. This paper investigates this problem in adversarial training, focusing on the generalization and optimization analyses of adversarial losses under weaker smoothness. We introduce Hölder continuity for the adversarial loss and propose a new concept of approximate Hölder smoothness, extending classical analyses from Lipschitz assumptions to a more general regime. A stability analysis for adversarial training is then developed, forming the theoretical basis for analyzing generalization and optimization. Under these conditions, we establish upper bounds on the generalization error and optimization gap of stochastic gradient descent for both convex and non-convex adversarial losses, and analyze the induced excess risk. These findings apply to a broad class of non-Lipschitz but Hölder smooth losses in deep learning, and offer theoretical insights into how optimization and generalization interact in adversarial training, contributing to a systematic understanding of robust overfitting. Furthermore, we empirically validate our theoretical findings using a practical example, demonstrating that the framework accurately captures generalization behavior under adversarial training.

Keywords Adversarial training · Robust generalization · Approximate Hölder smoothness · Generalization bound · Optimization gap

Editor: Yunwen Lei.

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Published online: 07 October 2026

1 Introduction

Deep neural networks (DNNs) have achieved remarkable success across various domains, including computer vision (Chen et al., 2025), natural language processing (Cai et al., 2025), and recommender systems (Lin et al., 2025). However, their vulnerability to adversarial perturbations poses significant challenges for deployment in safety applications (Zhang et al., 2023; Nguyen et al., 2025). These perturbations are typically small, human-imperceptible modifications intentionally crafted to induce erroneous predictions (Liu et al., 2024). Adversarial training (Chen et al., 2024; Hu et al., 2026) has emerged as a prominent approach for improving model robustness, which is commonly formulated as a minimax optimization problem:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\max_{\delta\in\Delta} \mathcal{L}\left(f_\theta(x+\delta), y\right) \right], \tag{1}$$

where θ denotes the model parameters, f_θ is a neural network parameterized by θ, δ is the adversarial perturbation, Δ is an admissible perturbation set, $\mathcal{L}(\cdot)$ is a loss function, $z = (x, y)$ is the input–output pair, $\mathcal{D}$ is the underlying data distribution, and $\mathbb{E}$ is the expectation. Despite its empirical success, adversarial training easily suffers from the robust overfitting issue (Rice et al., 2020; Li & Spratling, 2023b), where robust training accuracy continues to improve while robust test performance degrades, limiting robust generalization and motivating deeper theoretical investigations.

Understanding the robust generalization of adversarial training requires rigorous analysis of the gap between adversarial population risk and adversarial empirical risk. Existing studies have explored this problem under diverse theoretical frameworks. For example, Viallard et al. (2021) pioneered theoretical analysis of adversarial robustness by developing adversarial PAC-Bayes generalization bounds for majority vote classifiers under arbitrary attacks. Xiao et al. (2023) derived spectrally normalized generalization bounds for adversarial training under PAC-Bayes that match the tightness of standard bounds. Gu et al. (2024) proposed an adversarial data splitting for domain generalization and analyzed its generalization via VC dimension. Yin et al. (2019) analyzed adversarially robust generalization by proposing the notion of adversarial Rademacher complexity, revealing the intrinsic dimension dependence in adversarial settings. Along this line, Liu et al. (2023) proposed a stable adversarial learning algorithm that leveraged heterogeneous data sources, and established generalization guarantees via Rademacher complexity. Moreover, Zou and Liu (2023) employed the Rademacher complexity to analyze the generalization performance of adversarial contrastive learning. Maurya et al. (2025) provided a unified analysis of adversarial robustness by establishing tight upper and lower bounds on adversarial Rademacher complexity, yielding non-vacuous generalization guarantees.

Another line of work focuses on uniform stability analysis (Bousquet & Elisseeff, 2002), which characterizes the sensitivity of learning algorithms to perturbations in the training data. Hardt et al. (2016) established a theoretical foundation for the generalization of stochastic gradient descent (SGD) algorithms, while Bassily et al. (2020) further studied the uniform stability of SGD under nonsmooth convex losses. Deng et al. (2023) extended stability-based generalization analysis to asynchronous decentralized SGD. In adversarial training settings, uniform stability theory has also been investigated. Farnia and Ozdaglar

(2021) studied the stability of minimax optimization problems and proposed a gradient descent-based method to solve the inner maximization problem. Xing et al. (2021) derived the generalization error by using uniform argument stability. Xiao et al. (2022) established stability-based generalization bounds under the approximate gradient Lipschitz condition in adversarial training. More recently, Xiao et al. (2024) introduced a Moreau envelope function and analyzed uniform stability in adversarial training. Zhang et al. (2024) established generalization guarantees for adversarial training applied to two-layer networks with smooth activation functions. Tian and Mao (2025) analyzed the impact of training perturbations on generalization in adversarial settings, using the concept of expansiveness in perturbation maps. Wang et al. (2025) analyzed the on-average stability of SGD-based adversarial training and established data-dependent generalization bounds.

However, existing stability-based generalization analyses of adversarial training mainly rely on the assumption that the loss function or its gradient is Lipschitz continuous (Zühlke & Kudenko, 2025). While this assumption facilitates theoretical analysis, it imposes a strong global regularity requirement that significantly restricts the admissible class of loss functions. In particular, several commonly used loss functions in DNNs, such as q-norm hinge loss and q-norm regularization term ($1 \leq q \leq 2$), may fail to satisfy global Lipschitz continuity of the loss gradient (Lei & Ying, 2020; Virmaux & Scaman, 2018; Gao & Deng, 2024). To relax this stringent requirement, recent studies have explored weaker smoothness conditions for loss functions in DNNs. From a mathematical perspective, Lipschitz continuity is a special case of Hölder continuity, corresponding to a Hölder exponent equal to one, whereas Hölder continuity with an exponent strictly smaller than one characterizes a strictly larger class of functions and allows for weaker regularity of the gradient. In this direction, Lei and Ying (2020) showed that several commonly used loss functions satisfy Hölder continuity of the gradient rather than global Lipschitz continuity. Moreover, a growing body of work has begun to leverage Hölder continuity in the theoretical analysis of generalization bounds (Davidson & Dym, 2025; Deng et al., 2024). These results indicate that Hölder continuity provides a mathematically well-defined assumption for analyzing modern DNN models.

While these studies demonstrate the applicability of Hölder continuity for analyzing generalization in standard DNN settings, its theoretical implications for robust generalization in adversarial training remain largely unexplored. Particularly, it remains unclear how to analyze the optimization and generalization behavior of adversarial training under weak smoothness assumptions that are more general than the classical Lipschitz setting. Existing stability-based arguments, which play a central role in deriving generalization bounds for stochastic gradient methods, rely heavily on Lipschitz continuity and are not directly applicable to adversarial losses that exhibit only weaker smoothness. Again, the interplay among optimization error, generalization error, and excess risk under such weak smoothness conditions for convex and non-convex adversarial losses has not been fully understood.

This paper studies robust generalization of adversarial training in DNNs, emphasizing both optimization and generalization properties under general weak smoothness conditions. We introduce the Hölder continuity for the adversarial loss and further present a new concept of approximate Hölder smoothness for the loss gradient. Its stability properties for adversarial training are then established. Together, Hölder continuity and approximate Hölder smoothness form a unified term “Hölder-type conditions", which extends traditional Lipschitz-based analyses to a broader class of loss functions commonly encountered

in adversarial training. These conditions provide a foundation for investigating the optimization and generalization behavior of SGD in adversarial settings. Building on these conditions, upper bounds are rigorously established for the generalization error and the optimization gap for SGD under both convex and non-convex adversarial losses, while the corresponding excess risk is characterized. The findings provide a theoretical perspective on the interaction between optimization and generalization in adversarial training, offering new insights into understanding robust overfitting. An empirical study is conducted using a practical example that combines cross-entropy loss with a q-norm ($1 \leq q \leq 2$) regularization constraint. The results demonstrate that the Hölder-type loss can effectively capture the generalization behavior of models trained under adversarial perturbations, exhibiting strong consistency with both theoretical analyses and empirical observations. To facilitate understanding, Fig. 1 presents the core ideas and the interconnections among our theoretical results. Overall, the main contributions of this work are summarized as follows.

- A new notion of approximate Hölder smoothness is introduced, and its stability analysis is established, providing a foundation for analyzing robust generalization under weaker smoothness conditions in adversarial training.
- Under Hölder-type conditions, the generalization bound, optimization gap, and excess risk are rigorously derived for both convex and non-convex adversarial losses for SGD. These results apply to a broad class of non-Lipschitz but Hölder smooth losses in DNNs, and offer theoretical insights into how weak smoothness influences the interplay between optimization and generalization in adversarial training.
- A concrete loss with q-norm regularization ($1 \leq q \leq 2$) is provided as an example satisfying the Hölder-type conditions, which empirically validates the theoretical analysis under adversarial training.

The remainder of this paper is organized as follows. Section 2 introduces preliminaries on the adversarial training and some basic concepts. Sections 3 and 4 present our main

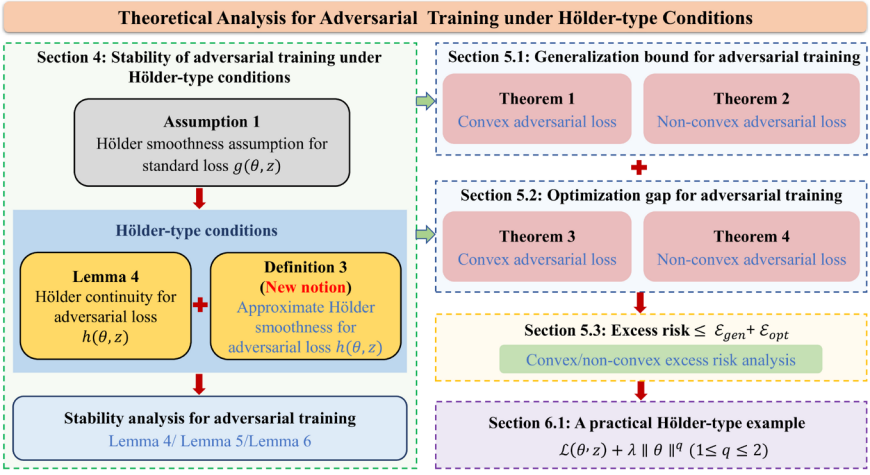


Fig. 1 The core ideas and interconnections among our theoretical results

theoretical results. Section 5 gives a practical example and analyzes the results, and Sect. 6 concludes the paper.

2 Preliminaries

2.1 Adversarial Training

DNNs are highly vulnerable to imperceptible perturbations in input data, which lead to substantial errors in model predictions (Liu et al., 2024; Zhao et al., 2025b). This vulnerability raises serious concerns about the reliability and safety of DNNs and has motivated extensive research on adversarial training. Since the seminal work (Goodfellow et al., 2015), numerous studies have explored adversarial training and its variants (Goyal et al., 2023; Waghela et al., 2024). By incorporating adversarially perturbed examples into the training process, adversarial training encourages models to learn decision boundaries that are resilient to worst-case perturbations. Recent advances have further expanded its applicability and effectiveness. For example, Zhou et al. (2023) proposed a latent boundary-guided adversarial training framework to achieve a tradeoff between standard accuracy and adversarial robustness. Lee et al. (2024) introduced a data-free framework that enables adversarial training by generating surrogate data in the absence of original training samples. Yuan et al. (2024) proposed an adaptive perturbation for adversarial attack, replacing the sign function in gradient-based attacks with a scaling factor and improving attack transferability. Zhao et al. (2025a) proposed multi-guided adversarial training to optimize decision boundaries. More works for adversarial training can be found in the survey (Qian et al., 2022).

Despite their substantial progress, recent studies have gradually shifted attention toward their generalization behavior and robust overfitting. Li and Spratling (2023a) investigated data augmentation strategies to simultaneously enhance both accuracy and robustness. Wang et al. (2024) explored benign overfitting in adversarial training. Angioni et al. (2025) proposed robustness-congruent adversarial training to alleviate performance regression in both accuracy and adversarial robustness during model updates.

2.2 Excess Risk Decomposition and Uniform Stability

Let $\mathcal{D}$ be an unknown data distribution over an instance space $\mathcal{Z}$. A training dataset is given by $S = \{z_1, z_2, \ldots, z_n\}$, where each $z_i = (x_i, y_i)$ is an input–output pair and is drawn i.i.d. from $\mathcal{D}$. Let $h(\theta, \cdot)$ be a loss function parameterized by θ, where θ represents all trainable parameters of a deep neural network. For example, a representative loss function is the cross-entropy loss for the classification tasks, which is commonly used in deep neural networks. Then the population risk and empirical risk are respectively defined as

$$R_{\mathcal{D}}(\theta) = \mathbb{E}_{z\sim\mathcal{D}}[h(\theta, z)] \quad \text{and} \quad R_S(\theta) = \frac{1}{n}\sum_{i=1}^{n} h(\theta, z_i).$$

Denote that θ^* and $\bar{\theta}$ are the minimizers of the population risk $R_{\mathcal{D}}(\theta)$ and the empirical risk $R_S(\theta)$, respectively. For the output $\hat{\theta} = \mathcal{A}(S)$ of a learning algorithm $\mathcal{A}$, the excess risk is decomposed as

$$R_{\mathcal{D}}(\hat{\theta}) - R_{\mathcal{D}}(\theta^*) = \underbrace{R_{\mathcal{D}}(\hat{\theta}) - R_S(\hat{\theta})}_{\varepsilon_{\text{gen}}} + \underbrace{R_S(\hat{\theta}) - R_S(\bar{\theta})}_{\varepsilon_{\text{opt}}} + \underbrace{R_S(\bar{\theta}) - R_S(\theta^*)}_{\leq 0} + \underbrace{R_S(\theta^*) - R_{\mathcal{D}}(\theta^*)}_{\mathbb{E}=0}. \tag{2}$$

This decomposition reveals that the excess risk is primarily determined by the generalization gap $\mathcal{E}_{\text{gen}}$ and the optimization gap $\mathcal{E}_{\text{opt}}$. The former measures how well the empirical loss approximates the population loss, while the latter quantifies the extent to which the algorithm succeeds in minimizing the empirical loss. The remaining two terms are either non-positive or have zero expectation and can therefore be safely ignored in the analysis.

A powerful tool for measuring generalization error is uniform stability (Bousquet & Elisseeff, 2002). Intuitively, an algorithm is stable if small perturbations in the training dataset do not significantly affect its output.

Definition 1 *(Uniform Stability* (Bousquet & Elisseeff, 2002)) A randomized algorithm $\mathcal{A}$ is ε-uniformly stable if for all datasets $S, S' \in \mathcal{Z}^n$ such that S and S' differ in at most one example, we have

$$\sup_z \mathbb{E}_{\mathcal{A}} \left[|h(\mathcal{A}(S), z) - h(\mathcal{A}(S'), z)| \right] \leq \varepsilon. \tag{3}$$

The following lemma establishes the connection between uniform stability and generalization in expectation.

Lemma 1 *(Generalization via Uniform Stability (Hardt et al., 2016)) If $\mathcal{A}$ be ε-uniformly stable, then*

$$|\mathcal{E}_{gen}| = |\mathbb{E}_{S,\mathcal{A}}[R_{\mathcal{D}}[\mathcal{A}(S)] - R_S[\mathcal{A}(S)]]| \leq \varepsilon\,, \tag{4}$$

2.3 Hölder Continuity and Its Property

Hölder continuity offers a more flexible and general notion of smoothness by relaxing the linear scaling constraint of Lipschitz functions, enabling more refined analyses of how the loss and its gradient respond to perturbations in the input or parameters.

Definition 2 *(Hölder smoothness* (Lei & Ying, 2020)) Let $L > 0$ and $m \in (0, 1]$. A function $g : \mathbb{R}^d \to \mathbb{R}$ is said to be (m, L)-Hölder smooth if for all $\theta_1, \theta_2 \in \mathbb{R}^d$, its gradient $\nabla g(\cdot)$ satisfies

$$\|\nabla g(\theta_1) - \nabla g(\theta_2)\| \leq L\|\theta_1 - \theta_2\|^m, \tag{5}$$

where $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^d$.
When $m = 1$, the condition in (5) reduces to the standard Lipschitz smoothness with Lipschitz constant L. Next, Hölder smoothness implies Hölder continuity of the function itself under mild boundedness conditions (Berger et al., 2020).

Lemma 2 *Let $\Theta \subset \mathbb{R}^d$ be a bounded set with diameter at most R, and let $g : \Theta \times \mathcal{X} \to \mathbb{R}$ be differentiable on θ. Suppose that for any fixed $x \in \mathcal{X}$, $g(\theta, x)$ is (m, L)-Hölder smooth with respect to θ for $m \in (0, 1]$ and $L > 0$. Then the following statements hold:*

(i) There exists a finite constant $M > 0$ such that $\|\nabla_\theta g(\theta, x)\| \leq M$ for all $\theta \in \Theta$ and $x \in \mathcal{X}$.
(ii) The function $g(\cdot, x)$ is (m, C)-Hölder continuous on Θ for all $x \in \mathcal{X}$, where $C = MR^{1-m}$.

A representative example satisfying Lemma 2 is the q-norm hinge loss $g(\theta, x) = (\max(0, 1 - y\langle\theta, x\rangle))^q$ $(q \in (1, 2])$, which is commonly used in the classification tasks.

3 Stability of Adversarial Training Under Hölder-Type Conditions

3.1 Hölder Smooth Assumptions for Adversarial Training

Adversarial loss. In adversarial training, the loss function is defined as

$$h(\theta, z) = \max_{\|\delta\|_p \leq \epsilon} g(\theta, x + \delta, y), \tag{6}$$

where δ is an adversarial perturbation constrained by the ℓ_p-norm $(p \geq 1)$ with radius ϵ , $g(\theta, x, y)$ is a standard loss (typically expressed as $\mathcal{L}(f_\theta(x), y)$ in Eq. (1)), $z = (x, y)$ is an input–output pair, and f_θ denotes a neural network parameterized by θ. For notational simplicity, we use $g(\theta, x)$ as shorthand for $g(\theta, x, y)$ in the subsequent analysis. The perturbation δ captures the worst-case loss within a local neighborhood of the input, enhancing the robustness of the objective against adversarial examples. A concrete example of the adversarial loss $h(\theta, z)$ in Eq. (6) is obtained when the function $g(\cdot)$ is the cross-entropy loss. To facilitate a rigorous stability and generalization analysis, we assume that g satisfies the following Hölder smoothness conditions:

Assumption 1 Assume that g is L_θ-Hölder smooth with respect to θ and L_z-Hölder smooth on x, i.e.,

$$\|\nabla_\theta g(\theta_1, x) - \nabla_\theta g(\theta_2, x)\| \leq L_\theta \|\theta_1 - \theta_2\|^m \text{ and } \|\nabla_\theta g(\theta, x_1) - \nabla_\theta g(\theta, x_2)\| \leq L_z \|x_1 - x_2\|_p^{m'}, \tag{7}$$

where $m, m' \in (0, 1]$, and $\|\cdot\|$ denotes the Euclidean norm.
A concrete example satisfying Assumption 1 is the q-norm hinge loss $g(\theta, x) = (\max(0, 1 - y\langle\theta, x\rangle))^q$, where $q \in (1, 2]$. By Lemma 2, it follows that $g(\theta, x)$ is C_θ -Hölder continuous with respect to θ, i.e.,

$$\|g(\theta_1, x) - g(\theta_2, x)\| \leq C_\theta \|\theta_1 - \theta_2\|^m. \tag{8}$$

To investigate the behavior of adversarial training under this setting, we first establish several key properties of the adversarial loss $h(\theta, z)$. Before proceeding, we state Danskin's theorem (Danskin, 2012).

Lemma 3 *(Danskin's Theorem (Danskin, 2012)) Let $\mathcal{X}$ be a compact set, and suppose that $f(\theta, x)$ is continuous in (θ, x) and continuously differentiable with respect to θ. Define $\phi(\theta) = \max_{x \in \mathcal{X}} f(\theta, x)$, and let $X^*(\theta) = \arg\max_{x \in \mathcal{X}} f(\theta, x)$. Then, the directional derivative of ϕ at θ in the direction v is given by*

$$\phi'(\theta; v) = \max_{x \in X^*(\theta)} v^\top \nabla_\theta f(\theta, x).$$

In particular, if $X^*(\theta) = \{x^*\}$, then ϕ is differentiable at θ, and

$$\nabla_\theta \phi(\theta) = \nabla_\theta f(\theta, x^*).$$

When multiple maximizers exist, we select an arbitrary $x^* \in X^*(\theta)$ and use $\nabla_\theta f(\theta, x^*)$ as the corresponding selected gradient. This convention will be used in the subsequent analysis of the adversarial loss. Building on Lemma 3 and Assumption 1, we next establish the following properties of the adversarial loss $h(\theta, z)$.

Lemma 4 *Assume that $h(\theta, z)$ is the adversarial loss defined in Eq. (6) with $g(\theta, x)$ meeting Assumption 1. Then for all $\theta_1, \theta_2 \in \Theta$ and $z \in \mathcal{Z}$, we have*

(i) $h(\theta, z)$ is (m, C_θ)-Hölder continuous with respect to θ, i.e.,

$$\|h(\theta_1, z) - h(\theta_2, z)\| \leq C_\theta \|\theta_1 - \theta_2\|^m.$$

(ii) For $i = 1, 2$, let $x_i \in \arg\max_{\|x' - x\|_p \leq \epsilon} g(\theta_i, x')$ be a selected maximizer, and define $d(\theta_i, z) = \nabla_\theta g(\theta_i, x_i)$ as the corresponding selected adversarial gradient. Then,

$$\|d(\theta_1, z) - d(\theta_2, z)\| \leq L_\theta \|\theta_1 - \theta_2\|^m + 2L_z \epsilon^{m'}.$$

Proof (i) Let

$$x_1 \in \arg\max_{\|x' - x\|_p \leq \epsilon} g(\theta_1, x'), \quad x_2 \in \arg\max_{\|x' - x\|_p \leq \epsilon} g(\theta_2, x').$$

By the optimality of x_1 and x_2, we have $g(\theta_1, x_1) \geq g(\theta_1, x_2)$ and $g(\theta_2, x_2) \geq g(\theta_2, x_1)$, which implies

$$\|h(\theta_1, z) - h(\theta_2, z)\| = |g(\theta_1, x_1) - g(\theta_2, x_2)| \leq \max\{|g(\theta_1, x_1) - g(\theta_2, x_1)|, |g(\theta_1, x_2) - g(\theta_2, x_2)|\}.$$

Applying the Hölder continuity of $g(\cdot, x)$ in Eq. (8) yields

$$\|h(\theta_1, z) - h(\theta_2, z)\| \leq C_\theta \|\theta_1 - \theta_2\|^m.$$

(ii) For $i = 1, 2$, let

$$x_i \in \arg\max_{\|x'-x\|_p \le \epsilon} g(\theta_i, x')$$

be a selected maximizer. Following the convention introduced after Lemma 3, we define the corresponding selected adversarial gradient as

$$d(\theta_i, z) = \nabla_\theta g(\theta_i, x_i).$$

Thus,

$$\begin{aligned}\|d(\theta_1, z) - d(\theta_2, z)\| &= \|\nabla_\theta g(\theta_1, x_1) - \nabla_\theta g(\theta_2, x_2)\| \\ &\le \|\nabla_\theta g(\theta_1, x_1) - \nabla_\theta g(\theta_2, x_1)\| + \|\nabla_\theta g(\theta_2, x_1) - \nabla_\theta g(\theta_2, x_2)\| \\ &\le L_\theta \|\theta_1 - \theta_2\|^m + L_z \|x_1 - x_2\|_p^{m'}.\end{aligned}$$

Since $\|x_1 - x\|_p \le \epsilon$ and $\|x_2 - x\|_p \le \epsilon$, it follows that $\|x_1 - x_2\|_p \le 2\epsilon$. Using $m' \in (0, 1]$, we obtain

$$\|d(\theta_1, z) - d(\theta_2, z)\| \le L_\theta \|\theta_1 - \theta_2\|^m + 2L_z \epsilon^{m'}.$$

This completes the proof. □

Remark 1 When $m = m' = 1$, Assumption 1 and Lemma 4 degenerate to the standard Lipschitz continuity or smoothness setting. Thus, our analysis generalizes the Lipschitz regime to the more general Hölder regime.

Remark 2 Since the adversarial loss $h(\theta, z)$ is defined through an inner maximization, it may be nonsmooth with respect to θ. Therefore, in Lemma 4(ii), we use $d(\theta, z)$ to explicitly denote the selected adversarial gradient associated with a selected maximizer of the inner problem. For notational simplicity, we use $\nabla_\theta h(\theta, z)$ in the subsequent analysis as a unified notation for $d(\theta, z)$. When the inner maximizer is unique, Danskin's theorem ensures that this notation coincides with the ordinary gradient of $h(\theta, z)$. This convention ensures the consistency of the subsequent analyses.

3.2 Approximate Hölder Smoothness and Basic Properties

We introduce a new notion of approximate Hölder smoothness inspired by Lemma 4, which relaxes the standard Hölder smoothness condition by allowing an additive term.

Definition 3 *(Approximate Hölder smoothness)* Let $\beta > 0$, $\eta > 0$, and $m \in (0, 1]$. A function $h(\theta, z)$ is said to be η-approximately β-Hölder smooth if for any $\theta_1, \theta_2 \in \Theta$ and $z \in \mathcal{Z}$,

$$\|\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z)\| \le \beta \|\theta_1 - \theta_2\|^m + \eta. \tag{9}$$

In Definition 3, η quantifies the degree of approximation in the smoothness condition of the adversarial loss function $h(\cdot)$. When $\eta = 0$, the definition reduces to the classical Hölder smoothness condition. This relaxed smoothness notion is particularly suitable for adversarial training, where exact gradient regularity may fail due to inner maximization.

It should be noted that similar relaxed assumptions have also been studied under generalized Lipschitz conditions (Zeng & Lei, 2026), where an additive constant is incorporated into the classical Lipschitz continuity condition in expectation. Under this generalized Lipschitz condition, Zeng and Lei (2026) developed a framework for studying the stability and generalization of biased stochastic gradient methods (BSGMs), with the relaxation accounting for the effects of bias and gradient estimators on algorithmic stability. In contrast, Definition 3 extends the standard Hölder smoothness condition by introducing an additive approximation term η into the adversarial loss gradient, which naturally arises from the inner maximization in adversarial training. Specifically, by Lemma 4(ii), the adversarial loss $h(\theta, z)$ is $2L_z\epsilon^{m'}$-approximately L_θ-Hölder smooth, corresponding to $\eta = 2L_z\epsilon^{m'}$.

Next, we present several fundamental properties of approximate Hölder smoothness, which play key roles in our subsequent stability and generalization analysis for adversarial training.

Lemma 5 *Suppose that $h(\theta, z)$ is η-approximately β-Hölder smooth and $m \in (0, 1]$. For any $\theta_1, \theta_2 \in \Theta$ and $z \in \mathcal{Z}$, we have*

(i) $h(\theta_1, z) - h(\theta_2, z) \le \nabla_\theta h(\theta_2, z)^T(\theta_1 - \theta_2) + \frac{\beta}{m+1}\|\theta_1 - \theta_2\|^{m+1} + \eta\|\theta_1 - \theta_2\|$;

(ii) If $h(\theta, z)$ is convex on θ for all $z \in \mathcal{Z}$, and $[\cdot]_+ = \max(0, \cdot)$, then

$$\langle \nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z),\, \theta_1 - \theta_2 \rangle \ge \frac{m}{\beta^{1/m}}\Big[\big(\|\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z)\| - \eta\big)_+\Big]^{1+1/m}.$$

Proof (i) Let $\tilde{\theta}$ be a point in the line segment of θ_1 and θ_2, we have $\tilde{\theta}(u) = \theta_2 + u(\theta_1 - \theta_2)$, then

$$\begin{aligned}
h(\theta_1, z) - h(\theta_2, z) &= \int_0^1 \langle \theta_1 - \theta_2, \nabla_\theta h(\tilde{\theta}(u), z)\rangle du \\
&= \int_0^1 \langle \theta_1 - \theta_2, \nabla_\theta h(\theta_2, z) + \nabla_\theta h(\tilde{\theta}(u), z) - \nabla_\theta h(\theta_2, z)\rangle du \\
&\le \langle \nabla_\theta h(\theta_2, z), \theta_1 - \theta_2\rangle \\
&\quad + \int_0^1 \|\theta_1 - \theta_2\| \cdot \|\nabla_\theta h(\tilde{\theta}(u), z) - \nabla_\theta h(\theta_2, z)\| du \\
&\le \langle \nabla_\theta h(\theta_2, z), \theta_1 - \theta_2\rangle + \int_0^1 \|\theta_1 - \theta_2\| \cdot [\beta\|\tilde{\theta}(u) - \theta_2\|^m + \eta] du \\
&= \langle \nabla_\theta h(\theta_2, z), \theta_1 - \theta_2\rangle + \int_0^1 \|\theta_1 - \theta_2\| \cdot [\beta u^m \|\theta_1 - \theta_2\|^m + \eta] du \\
&= \nabla_\theta h(\theta_2, z)^T(\theta_1 - \theta_2) + \frac{\beta}{m+1}\|\theta_1 - \theta_2\|^{m+1} + \eta\|\theta_1 - \theta_2\|.
\end{aligned}$$

(ii) By (i), we have

$$h(\theta_1, z) \leq h(\theta_2, z) + \nabla_\theta h(\theta_2, z)^T(\theta_1 - \theta_2) + \frac{\beta}{m+1}\|\theta_1 - \theta_2\|^{m+1} + \eta\|\theta_1 - \theta_2\|.$$

Let θ^* be a minimizer of $h(\theta, z)$, $t = \|\theta_1 - \theta_2\|$ and $\upsilon = (\theta_1 - \theta_2)/\|\theta_1 - \theta_2\|$, then

$$\begin{aligned}
h(\theta^*, z) &= \inf_{\theta_1} h(\theta_1, z) \\
&\leq \inf_{\theta_1}\left(h(\theta_2, z) + \nabla_\theta h(\theta_2, z)^T(\theta_1 - \theta_2) + \frac{\beta}{m+1}\|\theta_1 - \theta_2\|^{m+1} + \eta\|\theta_1 - \theta_2\|\right) \\
&= \inf_{\|\upsilon\|=1}\inf_{t\geq 0}\left(h(\theta_2, z) + t\nabla_\theta h(\theta_2, z)^T\upsilon + \frac{\beta}{m+1}t^{m+1} + \eta t\right) \\
&= \inf_{t\geq 0}\left(h(\theta_2, z) - t\|\nabla_\theta h(\theta_2, z)\| + \frac{\beta}{m+1}t^{m+1} + \eta t\right) \\
&= h(\theta_2, z) + \inf_{t\geq 0}\left(-t(\|\nabla_\theta h(\theta_2, z)\| - \eta) + \frac{\beta}{m+1}t^{m+1}\right).
\end{aligned}$$

If $\|\nabla_\theta h(\theta_2, z)\| - \eta \leq 0$, then the function is optimized when $t = 0$, i.e.,

$$h(\theta_2, z) + \inf_{t\geq 0}\left(-t(\|\nabla_\theta h(\theta_2, z)\| - \eta) + \frac{\beta}{m+1}t^{m+1}\right) = h(\theta_2, z).$$

If $\|\nabla_\theta h(\theta_2, z)\| - \eta \geq 0$, then

$$\begin{aligned}
&h(\theta_2, z) + \inf_{t\geq 0}\left(-t(\|\nabla_\theta h(\theta_2, z)\| - \eta) + \frac{\beta}{m+1}t^{m+1}\right) \\
&= h(\theta_2, z) - \frac{m}{m+1}\cdot\frac{1}{\beta^{\frac{1}{m}}}[\|\nabla_\theta h(\theta_2, z)\| - \eta]^{\frac{1}{m}+1}.
\end{aligned}$$

Therefore, we obtain that

$$h(\theta^*, z) - h(\theta, z) \leq -\frac{m}{m+1}\cdot\frac{1}{\beta^{\frac{1}{m}}}\left[[\|\nabla_\theta h(\theta, z)\| - \eta]_+\right]^{1+\frac{1}{m}}.$$

Define $h_1(w, z) = h(w, z) - \nabla_\theta h(\theta_1, z)^T w$ and $h_2(w, z) = h(w, z) - \nabla_\theta h(\theta_2, z)^T w$. According to Definition 3, $h_1(w, z)$ and $h_2(w, z)$ are both η-approximately β-Hölder smooth on w.

Besides, when $(w, z) = (\theta_1, z)$ minimizes $h_1(w, z)$, we have

$$\begin{aligned}
h_1(\theta_2, z) - h_1(\theta_1, z) &= h(\theta_2, z) - h(\theta_1, z) - \nabla_\theta h(\theta_1, z)^T(\theta_2 - \theta_1) \\
&\geq \frac{m}{m+1}\cdot\frac{1}{\beta^{\frac{1}{m}}}\left[[\|\nabla_\theta h_1(\theta_2, z)\| - \eta]_+\right]^{1+\frac{1}{m}} \\
&= \frac{m}{m+1}\cdot\frac{1}{\beta^{\frac{1}{m}}}\left[[\|\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z)\| - \eta]_+\right]^{1+\frac{1}{m}}.
\end{aligned} \tag{10}$$

Similarly, when $(w, z) = (\theta_2, z)$ minimizes $h_2(w, z)$, we have

$$h_2(\theta_1, z) - h_2(\theta_2, z) = h(\theta_1, z) - h(\theta_2, z) - \nabla_\theta h(\theta_2, z)^T(\theta_1 - \theta_2) \geq \frac{m}{m+1} \cdot \frac{1}{\beta^{\frac{1}{m}}} [[\|\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z)\| - \eta]_+]^{1+\frac{1}{m}}. \tag{11}$$

Taking the summation of Eqs. (10) and (11), we have

$$\begin{aligned}\langle \nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z), \theta_1 - \theta_2 \rangle &\geq \frac{2m}{m+1} \\ &\quad \cdot \frac{1}{\beta^{\frac{1}{m}}} [[\|\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z)\| - \eta]_+]^{1+\frac{1}{m}} \\ &\geq m \cdot \frac{1}{\beta^{\frac{1}{m}}} [[\|\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z)\| - \eta]_+]^{1+\frac{1}{m}}.\end{aligned}$$

This completes the proof. □

As a concrete example, the q-norm hinge loss $g(\theta, x, y) = (\max(0, 1 - y\langle \theta, x \rangle))^q$ ($q \in (1, 2]$) satisfies Assumption 1 with Hölder exponent $m = q - 1$. Consequently, by Lemma 4, the corresponding adversarial loss $h(\theta, z) = \max_{\|\delta\|_p \leq \epsilon} g(\theta, x + \delta, y)$ satisfies the η-approximately β -Hölder smoothness condition stated in Definition 3. Since Lemma 5 follows directly from Definition 3 and Lemma 4, the above adversarial loss $h(\theta, z)$ satisfies Lemma 5.

Next, we study the behavior of SGD applied to the adversarial loss under the approximate Hölder smoothness condition. Within the uniform stability framework, a central task is to quantify the deviation between two SGD trajectories generated from neighboring datasets that differ by a single sample. Let θ_1^T and θ_2^T denote the parameters obtained after T iterations of SGD on datasets S and S', respectively. Bounding the distance $\|\theta_1^T - \theta_2^T\|$ under approximate Hölder smoothness forms the basis of the recursive stability analysis that follows. We consider SGD for minimizing the adversarial loss. At iteration t, the update rule is given by

$$\theta^{t+1} = \theta^t - \alpha_t \nabla_\theta h(\theta^t, z_{i_t}), \tag{12}$$

where α_t is the step size at iteration t, and z_{i_t} is a training sample selected uniformly at random from the dataset. For notational convenience, we define the SGD update mapping as

$$G_{\alpha,z}(\theta) = \theta - \alpha \nabla_\theta h(\theta, z). \tag{13}$$

Lemma 6 *Let $m \in (0, 1]$, and $\Theta \subset \mathbb{R}^d$ be a bounded set with diameter at most R, i.e., $\|\theta_1 - \theta_2\| \leq R$ for all $\theta_1, \theta_2 \in \Theta$. Assume that the adversarial loss $h(\theta, z)$ is η-approximately β-Hölder smooth on $\Theta \times \mathcal{Z}$. Then, for any $\theta_1, \theta_2 \in \Theta$ and $z \in \mathcal{Z}$, the following statements hold:*

(i) $\|G_{\alpha,z}(\theta_1) - G_{\alpha,z}(\theta_2)\| \leq \|\theta_1 - \theta_2\| + \alpha\beta\|\theta_1 - \theta_2\|^m + \alpha\eta$;
(ii) Further, if $h(\theta, z)$ is convex in θ for all $z \in \mathcal{Z}$, then for any step size $\alpha \leq 1/C$ and

$$C = \left(\frac{\beta^{\frac{1}{m}}}{m}\right)^{\frac{2m}{m+1}},$$

$$\|G_{\alpha,z}(\theta_1) - G_{\alpha,z}(\theta_2)\| \le \|\theta_1 - \theta_2\| + \sqrt{\frac{2(1-m)}{1+m}\left(C\alpha^{\frac{2}{m+1}}\right)^{\frac{1+m}{1-m}}} + \sqrt{2}\alpha\eta.$$

Proof (i) Since $h(\theta, z)$ is η-approximately β-Hölder smooth, we have

$$\|\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z)\| \le \beta\|\theta_1 - \theta_2\|^m + \eta. \tag{14}$$

Using $G_{\alpha,z}(\theta) = \theta - \alpha\nabla_\theta h(\theta, z)$, we obtain

$$\begin{aligned}\|G_{\alpha,z}(\theta_1) - G_{\alpha,z}(\theta_2)\| &= \|\theta_1 - \theta_2 - \alpha(\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z))\| \\ &\le \|\theta_1 - \theta_2\| + \alpha\|\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z)\| \\ &\le \|\theta_1 - \theta_2\| + \alpha\beta\|\theta_1 - \theta_2\|^m + \alpha\eta,\end{aligned}$$

which proves part (i).

(ii) Assume further that $h(\theta, z)$ is convex on θ for all $z \in \mathcal{Z}$. Let

$$t = \|\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z)\|.$$

Case 1: $t \le \eta$ **.** By the triangle inequality, we directly have

$$\begin{aligned}\|G_{\alpha,z}(\theta_1) - G_{\alpha,z}(\theta_2)\| &\le \|\theta_1 - \theta_2\| + \alpha\eta \\ &\le \|\theta_1 - \theta_2\| + \sqrt{\frac{2(1-m)}{1+m}\left(C\alpha^{\frac{2}{m+1}}\right)^{\frac{1+m}{1-m}}} + \sqrt{2}\alpha\eta.\end{aligned}$$

Case 2: $t > \eta$ **.** Expanding the squared norm yields

$$\begin{aligned}\|G_{\alpha,z}(\theta_1) - G_{\alpha,z}(\theta_2)\|^2 &= \|\theta_1 - \theta_2\|^2 - 2\alpha\langle\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z), \theta_1 - \theta_2\rangle + \alpha^2 t^2 \\ &= \|\theta_1 - \theta_2\|^2 - 2\alpha\langle\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z), \theta_1 - \theta_2\rangle \\ &\quad + \alpha^2[(t-\eta) + \eta]^2 \\ &\le \|\theta_1 - \theta_2\|^2 - 2\alpha\langle\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z), \theta_1 - \theta_2\rangle \\ &\quad + 2\alpha^2(t-\eta)^2 + 2\alpha^2\eta^2.\end{aligned} \tag{15}$$

By Lemma 5(ii), the convexity of h implies

$$\langle\nabla_\theta h(\theta_1, z) - \nabla_\theta h(\theta_2, z), \theta_1 - \theta_2\rangle \ge \frac{2m}{m+1} \cdot \frac{1}{\beta^{\frac{1}{m}}}(t-\eta)^{1+\frac{1}{m}},$$

then, we have

$$(t-\eta)^2 \leq \left[\frac{(m+1)\beta^{\frac{1}{m}}}{2m}\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle\right]^{\frac{2m}{m+1}}.$$

Then

$$\begin{aligned}2\alpha^2(t-\eta)^2 &\leq 2\alpha^2\left[\frac{(m+1)\beta^{\frac{1}{m}}}{2m}\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle\right]^{\frac{2m}{m+1}}\\ &\leq 2\alpha^2\left[\frac{\beta^{\frac{1}{m}}}{m}\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle\right]^{\frac{2m}{m+1}}\\ &\triangleq 2\alpha^2 C[\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle]^{\frac{2m}{m+1}},\end{aligned} \tag{16}$$

where $C=\left(\frac{\beta^{\frac{1}{m}}}{m}\right)^{\frac{2m}{m+1}}$.

Applying Young's inequality with $p=\frac{1+m}{2m}$, $q=\frac{1+m}{1-m}$, we obtain

$$\begin{aligned}&\alpha^2 C[\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle]^{\frac{2m}{m+1}}\\ =&(\alpha\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle)^{\frac{2m}{m+1}}\cdot C\alpha^{\frac{2}{m+1}}\\ \leq&\frac{2m}{1+m}\left[(\alpha\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle)^{\frac{2m}{m+1}}\right]^{\frac{1+m}{2m}}+\frac{1-m}{1+m}\left(C\alpha^{\frac{2}{m+1}}\right)^{\frac{1+m}{1-m}}\\ \leq&\alpha\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle+\frac{1-m}{1+m}\left(C\alpha^{\frac{2}{m+1}}\right)^{\frac{1+m}{1-m}}.\end{aligned}$$

Substituting this into Eq. (16), we obtain

$$2\alpha^2(t-\eta)^2 \leq 2\alpha\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle+\frac{2(1-m)}{1+m}\left(C\alpha^{\frac{2}{m+1}}\right)^{\frac{1+m}{1-m}}.$$

Then Eq. (15) yields

$$\begin{aligned}\|G_{\alpha,z}(\theta_1)-G_{\alpha,z}(\theta_2)\|^2 &\leq \|\theta_1-\theta_2\|^2-2\alpha\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle\\ &\quad+2\alpha^2(t-\eta)^2+2\alpha^2\eta^2\\ &\leq \|\theta_1-\theta_2\|^2-2\alpha\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle\\ &\quad+2\alpha\langle\nabla_\theta h(\theta_1,z)-\nabla_\theta h(\theta_2,z),\theta_1-\theta_2\rangle\\ &\quad+\frac{2(1-m)}{1+m}\left(C\alpha^{\frac{2}{m+1}}\right)^{\frac{1+m}{1-m}}+2\alpha^2\eta^2\\ &=\|\theta_1-\theta_2\|^2+\frac{2(1-m)}{1+m}\left(C\alpha^{\frac{2}{m+1}}\right)^{\frac{1+m}{1-m}}+2\alpha^2\eta^2.\end{aligned}$$

Then

$$\|G_{\alpha,z}(\theta_1) - G_{\alpha,z}(\theta_2)\| = \sqrt{\|\theta_1 - \theta_2\|^2 + \frac{2(1-m)}{1+m}\left(C\alpha^{\frac{2}{m+1}}\right)^{\frac{1+m}{1-m}} + 2\alpha^2\eta^2}$$
$$\leq \|\theta_1 - \theta_2\| + \sqrt{\frac{2(1-m)}{1+m}\left(C\alpha^{\frac{2}{m+1}}\right)^{\frac{1+m}{1-m}}} + \sqrt{2}\alpha\eta.$$

Therefore $\|G_{\alpha,z}(\theta_1) - G_{\alpha,z}(\theta_2)\| \leq \|\theta_1 - \theta_2\| + \sqrt{\frac{2(1-m)}{1+m}\left(C\alpha^{\frac{2}{m+1}}\right)^{\frac{1+m}{1-m}}} + \sqrt{2}\alpha\eta.$ This completes the proof. □

Lemma 6 establishes a one-step recursive bound on the parameter deviation $\|\theta_1^t - \theta_2^t\|$ between two SGD trajectories at consecutive iterations when optimizing the adversarial loss under the approximate Hölder smoothness condition. By iteratively applying this recursion, we can control the cumulative deviation $\|\theta_1^T - \theta_2^T\|$ after T iterations. This recursive characterization constitutes the key technical ingredient in our uniform stability analysis. Building upon this result, we derive explicit uniform stability-based generalization bounds for adversarial training in both convex and non-convex loss settings.

4 Generalization and Optimization Analysis for Adversarial Training

4.1 Generalization Bounds Under Hölder-Type Conditions

Based on the preceding analysis, we now derive generalization bounds for adversarial training under the Hölder-type conditions in Sect. 3. Our analysis covers both convex and non-convex adversarial loss settings within the uniform stability framework.

i) **Convex Adversarial Loss**

Theorem 1 *(Generalization bounds under convex adversarial loss) Let $m, m' \in (0,1]$, and $\Theta \subset \mathbb{R}^d$ be a bounded set with diameter at most R. Assume that $h(\theta, z)$ is convex on θ, C_θ -Hölder continuous, and η-approximately β-Hölder smooth on Θ for all $z \in \mathcal{Z}$. Suppose that SGD is run for T iterations with step sizes $\alpha_t \leq 1/C$, where $C = \left(\frac{\beta^{\frac{1}{m}}}{m}\right)^{\frac{2m}{m+1}}$. Then, it follows that*

$$\mathcal{E}_{\text{gen}} := \mathbb{E}\left[R_{\mathcal{D}}(\theta^{T+1}) - R_S(\theta^{T+1})\right]$$
$$\leq C_\theta\left(\sqrt{2}\eta + \sqrt{\frac{2C(1-m)}{1+m}} + \frac{L_\theta R^m + 2L_z\epsilon^{m'}}{n}\right)^m \left(\sum_{t=1}^{T}\alpha_t\right)^m. \tag{17}$$

Proof Let S and S' be two sample sets of size n that differ in exactly one example. Let $\{\theta_1^t\}_{t=1}^{T+1}$ and $\{\theta_2^t\}_{t=1}^{T+1}$ denote the SGD iterates generated on S and S', respectively. Define $\delta_t := \|\theta_1^t - \theta_2^t\|$.

Fixing any $z \in \mathcal{Z}$, by the C_θ-Hölder continuity of $h(\theta, z)$ with respect to θ and Lemma 4(i), we obtain

$$\mathbb{E}\left|h(\theta_1^{T+1}, z) - h(\theta_2^{T+1}, z)\right| \leq C_\theta \mathbb{E}\left[\delta_{T+1}^m\right] \leq C_\theta \left(\mathbb{E}[\delta_{T+1}]\right)^m \triangleq \varepsilon, \tag{18}$$

where the last inequality follows from Jensen's inequality since $m \in (0, 1]$.

At iteration t, with probability $1 - \frac{1}{n}$, SGD selects the same training example from S and S'. In this case, Lemma 6(ii) implies

$$\begin{aligned}
\mathbb{E}[\delta_{t+1}] &\leq \mathbb{E}[\delta_t] + \sqrt{\frac{2(1-m)}{1+m}\left(C\alpha_t^{\frac{2}{m+1}}\right)^{\frac{1+m}{1-m}}} + \sqrt{2}\alpha_t\eta \\
&= \mathbb{E}[\delta_t] + \sqrt{\frac{2(1-m)}{1+m}} C^{\frac{1+m}{2(1-m)}} \alpha_t^{\frac{1}{1-m}} + \sqrt{2}\alpha_t\eta \\
&= \mathbb{E}[\delta_t] + \alpha_t \left(\sqrt{\frac{2(1-m)}{1+m}} C^{\frac{1+m}{2(1-m)}} \alpha_t^{\frac{m}{1-m}} + \sqrt{2}\eta\right) \\
&\leq \mathbb{E}[\delta_t] + \alpha_t \left(\sqrt{\frac{2(1-m)}{1+m}} C^{\frac{1+m}{2(1-m)}} \left(\frac{1}{C}\right)^{\frac{m}{1-m}} + \sqrt{2}\eta\right) \\
&= \mathbb{E}[\delta_t] + \alpha_t \left(\sqrt{\frac{2C(1-m)}{1+m}} + \sqrt{2}\eta\right).
\end{aligned} \tag{19}$$

With probability $\frac{1}{n}$, the selected examples differ. By Assumption 1, for any $x_1, x_2 \in \mathcal{X}$,

$$\|\nabla_\theta g(\theta, x_1) - \nabla_\theta g(\theta, x_2)\| \leq L_z \|x_1 - x_2\|_p^{m'}.$$

Since

$$h(\theta, z) = \max_{\|\delta\|_p \leq \epsilon} g(\theta, x + \delta),$$

any subgradient $\nabla_\theta h(\theta, z)$ can be expressed as $\nabla_\theta g(\theta, x + \delta^*)$ for some maximizer δ^*. Thus, for two examples $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$, we have

$$\|\nabla_\theta h(\theta, z_1) - \nabla_\theta h(\theta, z_2)\| \leq L_z \|x_1 - x_2\|_p^{m'} \leq L_z (\|x_1 - x\|_p + \|x - x_2\|_p)^{m'} \leq 2L_z \epsilon^{m'}.$$

Using the approximate Hölder smoothness of $h(\theta, z)$ with respect to θ (with $\beta = L_\theta$ and $\eta = 2L_z\epsilon^{m'}$), we obtain

$$\begin{aligned}\delta_{t+1} &\leq \delta_t + \alpha_t(\left\|\nabla_\theta h(\theta_1^t, z) - \nabla_\theta h(\theta_2^t, z')\right\|)\\ &\leq \delta_t + \alpha_t(\left\|\nabla_\theta h(\theta_1^t, z) - \nabla_\theta h(\theta_2^t, z)\right\| + \left\|\nabla_\theta h(\theta_2^t; z) - \nabla_\theta h(\theta_2^t, z')\right\|)\\ &\leq \delta_t + \alpha_t(\beta\left\|\theta_1^t - \theta_2^t\right\|^m + \eta + 2L_z\epsilon^{m'})\\ &\leq \delta_t + \alpha_t(L_\theta R^m + 4L_z\epsilon^{m'}).\end{aligned}$$

Taking expectation yields

$$\mathbb{E}[\delta_{t+1}] \leq \mathbb{E}[\delta_t] + \alpha_t(L_\theta R^m + 4L_z\epsilon^{m'}). \tag{20}$$

Combining (19) and (20) via a probability-weighted average, we obtain

$$\begin{aligned}\mathbb{E}[\delta_{t+1}] &\leq \left(1 - \frac{1}{n}\right)\left[\mathbb{E}[\delta_t] + \alpha_t\left(\sqrt{\frac{2C(1-m)}{1+m}} + \sqrt{2}\eta\right)\right] + \frac{1}{n}(\mathbb{E}[\delta_t] + \alpha_t(L_\theta R^m\\ &\quad + 4L_z\epsilon^{m'}))\\ &\leq \mathbb{E}[\delta_t] + \left(\sqrt{2}\eta + \sqrt{\frac{2C(1-m)}{1+m}} + \frac{L_\theta R^m + 2L_z\epsilon^{m'}}{n}\right)\alpha_t.\end{aligned}$$

Unrolling the recursion over $t = 1, \ldots, T$ gives

$$\mathbb{E}[\delta_{T+1}] \leq \left(\sqrt{2}\eta + \sqrt{\frac{2C(1-m)}{1+m}} + \frac{L_\theta R^m + 2L_z\epsilon^{m'}}{n}\right)\sum_{t=1}^{T}\alpha_t.$$

Substituting this bound into (18) and using the uniform stability-based generalization in Lemma 1 yields

$$\mathcal{E}_{\text{gen}} \leq C_\theta\left(\sqrt{2}\eta + \sqrt{\frac{2C(1-m)}{1+m}} + \frac{L_\theta R^m + 2L_z\epsilon^{m'}}{n}\right)^m\left(\sum_{t=1}^{T}\alpha_t\right)^m \triangleq \varepsilon.$$

This completes the proof. □

Given that the adversarial loss $h(\theta, z)$ is $2L_z\epsilon^{m'}$-approximately L_θ-Hölder smooth, setting $\eta = 2L_z\epsilon^{m'}$ in Eq. (17) yields an explicit generalization bound for adversarial training. When the sample size n is sufficiently large, the term $\frac{L_\theta R^m + 2L_z\epsilon^{m'}}{n}$ becomes negligible. Moreover, under a fixed learning rate $\alpha_t = \alpha$, Theorem 1 implies

$$\mathcal{E}_{\text{gen}} = \mathcal{O}\left(\left(\epsilon^{m'} + \sqrt{\frac{1-m}{1+m}}\right)^m T^m\right),$$

which explicitly characterizes the dependence of the generalization error on both the adversarial perturbation magnitude and the number of SGD iterations.

Remark 3 *(Connection to Lipschitz-based stability analysis)* Our analysis can be viewed as a natural extension of the stability analysis in (Xiao et al., 2022), where the adversarial loss is assumed to be Lipschitz smooth, yielding

$$\mathcal{E}_{\text{gen}} \leq \mathcal{O}(\epsilon T).$$

Under the weaker assumption of Hölder smoothness, our analysis establishes

$$\mathcal{E}_{\text{gen}} = \mathcal{O}\left(\left(\epsilon^{m'} + \sqrt{\frac{1-m}{1+m}}\right)^{m} T^{m}\right), \qquad m, m' \in (0,1]. \tag{21}$$

When $m = m' = 1$, the second term vanishes and the above bound reduces to the Lipschitz-based bound, thereby ensuring consistency with prior work. Also, for $0 < m, m' < 1$, our bound applies to a broader class of losses that may not be Lipschitz smooth but still satisfy Hölder smooth, providing a more flexible form of adversarial training in convex settings.

ii) **Non-convex adversarial loss**

We extend the above stability-based generalization analysis to adversarial training under non-convex losses.

Theorem 2 *(Generalization bounds under non-convex adversarial loss) Let $m, m' \in (0,1]$, and $\Theta \subset \mathbb{R}^d$ be a bounded set with diameter at most R. Assume that the adversarial loss $h(\theta, z)$ is C_θ-Hölder continuous and η-approximately β-Hölder smooth on $\theta \in \Theta$ for all $z \in \mathcal{Z}$, and its inner loss $g(\cdot)$ satisfies $0 \leq g(\theta, x) \leq B$. Suppose that SGD is run for T iterations with diminishing step sizes $\alpha_t \leq 1/\beta t$. Then, it follows that*

$$\mathcal{E}_{\text{gen}} \leq \frac{B}{n} + C_\theta \left(\eta + \beta R^m + \frac{L_\theta R^m + 2L_z \epsilon^{m'}}{n}\right)^{m} \left(\sum_{t=t_0}^{T} \alpha_t\right)^{m}. \tag{22}$$

Proof Let S and S' be two sample sets of size n that differ in exactly one example. Let $\{\theta_1^t\}_{t=1}^{T+1}$ and $\{\theta_2^t\}_{t=1}^{T+1}$ denote the SGD trajectories generated on S and S', respectively. Define $\delta_t = \|\theta_1^t - \theta_2^t\|$.

Let $t_0 \in \{1, \ldots, n\}$ denote the first iteration at which the two SGD runs select different samples, so that $\delta_{t_0} = 0$ but $\delta_{t_0+1} > 0$. Let $\mathcal{E} = 1\left[\delta_{t_0} = 0\right]$ be the event that the trajectories remain identical up to iteration t_0 and $\mathcal{E}^c$ be its complementary set.

By conditioning on whether the event occurs before iteration t_0, the generalization error can be decomposed as

$$\begin{aligned}
\mathcal{E}_{\text{gen}} \leq & \mathbb{P}\left\{\mathcal{E}^c\right\} \mathbb{E}\left[\left|h\left(\theta_{T+1}^1, z\right) - h\left(\theta_{T+1}^2, z\right)\right| \mid \mathcal{E}^c\right] + \mathbb{P}\{\mathcal{E}\} \mathbb{E}\left[\left|h\left(\theta_{T+1}^1, z\right) - h\left(\theta_{T+1}^2, z\right)\right| \mid \mathcal{E}\right] \\
\leq & \mathbb{P}\left\{\mathcal{E}^c\right\} \cdot \sup_{\theta, z} h(\theta, z) + C_\theta \cdot \mathbb{E}\left[\left\|\theta_{T+1}^1 - \theta_{T+1}^2\right\|^{m} \mid \mathcal{E}\right] \\
\leq & \frac{t_0}{n} B + C_\theta \mathbb{E}[\delta_{T+1}^m \mid \delta_{t_0} = 0] \\
\leq & \frac{t_0}{n} B + C_\theta (\mathbb{E}[\delta_{T+1} \mid \delta_{t_0} = 0])^m.
\end{aligned} \tag{23}$$

Since $\Delta_t = \mathbb{E}[\delta_t \mid \delta_{t_0} = 0]$, under the event $\mathcal{E}$, the two trajectories coincide until iteration t_0. Using the C_θ-Hölder of the adversarial loss $h(\theta, z)$ on θ, the conditional expectation in (23) can be upper bounded by $\mathbb{E}[\delta^m_{T+1} \mid \delta_{t_0} = 0]$.

At each iteration $t \geq t_0$, with probability $1 - \frac{1}{n}$, SGD selects the same training example in both datasets, while with probability $\frac{1}{n}$, the selected examples differ. Applying Lemma 6(i) and taking conditional expectation yields

$$\begin{aligned}\Delta_{t+1} &\leq (1-\frac{1}{n})(\Delta_t + \alpha_t(\eta + \beta R^m)) + \frac{1}{n}(\Delta_t + \alpha_t\left(L_\theta R^m + 4L_z\epsilon^{m'}\right)) \\ &= \Delta_t + (1-\frac{1}{n})\alpha_t(\eta + \beta R^m) + \frac{1}{n}\alpha_t\left(L_\theta R^m + 4L_z\epsilon^{m'}\right) \\ &\leq \Delta_t + (\eta + \beta R^m + \frac{L_\theta R^m + 2L_z\epsilon^{m'}}{n})\alpha_t.\end{aligned} \tag{24}$$

Since $\Delta_{t_0} = 0$, we can recursively unwind the above inequality from iteration $T+1$ down to $t_0 + 1$. This gives

$$\begin{aligned}\Delta_{T+1} &\leq \Delta_{t_0} + (\eta + \beta R^m + \frac{L_\theta R^m + 2L_z\epsilon^{m'}}{n})\sum_{t=t_0}^{T}\alpha_t \\ &= (\eta + \beta R^m + \frac{L_\theta R^m + 2L_z\epsilon^{m'}}{n})\sum_{t=t_0}^{T}\alpha_t.\end{aligned}$$

Plugging this bound into Eq. (23), we get

$$\mathcal{E}_{\text{gen}} \leq \frac{Bt_0}{n} + C_\theta\left((\eta + \beta R^m + \frac{L_\theta R^m + 2L_z\epsilon^{m'}}{n})\sum_{t=t_0}^{T}\alpha_t\right)^m.$$

If we simply set $t_0 = 1$, then

$$\mathcal{E}_{\text{gen}} \leq \frac{B}{n} + C_\theta\left(\eta + \beta R^m + \frac{L_\theta R^m + 2L_z\epsilon^{m'}}{n}\right)^m\left(\sum_{t=t_0}^{T}\alpha_t\right)^m.$$

This completes the proof. □

The non-convex generalization bound in Theorem 2 admits a similar interpretation to that in the convex setting.This reflects the fact that, within the uniform stability framework, the analysis is largely insensitive to whether the loss function is convex or non-convex. Overall, our analysis extends existing uniform stability-based generalization bounds to the Hölder-type conditions for both convex and non-convex adversarial losses, thereby broadening their applicability beyond the classical Lipschitz setting. This theoretical insight offers a unified perspective for understanding robust generalization in adversarial training.

4.2 Optimization Gap Under Hölder-Type Conditions

According to the excess risk decomposition in Eq. (2), we now analyze the optimization error bound of SGD under the more general Hölder-type conditions for both convex and non-convex adversarial losses.

i) **Convex adversarial loss**

Theorem 3 *(Optimization error under convex adversarial loss) Let $m, m' \in (0,1]$, and $\Theta \subset \mathbb{R}^d$ be a bounded set with diameter at most R. Assume that $h(\theta, z)$ is convex on θ, C_θ -Hölder continuous, and η-approximately β-Hölder smooth on Θ for all $z \in \mathcal{Z}$. Let $\bar{\theta}$ denote an optimal solution and $D = \|\theta^1 - \bar{\theta}\|$, where θ^1 is the initialization of SGD. Suppose that SGD is run for T iterations with step sizes $\{\alpha_t\}_{t=1}^T$. Then, there exists an $k \leq T$ such that*

$$\mathcal{E}_{\text{opt}}(\theta^k) \leq \frac{D^2 + \left(L_\theta R^m + 2L_z \epsilon^{m'}\right)^2 \sum_{t=1}^T \alpha_t^2}{2\sum_{t=1}^T \alpha_t}. \tag{25}$$

Proof Since $h(\theta, z)$ is convex with respect to θ for any fixed z, the empirical risk $R_S(\theta) = \frac{1}{n}\sum_{i=1}^n h(\theta, z_i)$ is also convex. For convenience of notation, let $f(\theta) = R_S(\theta)$. Therefore, for any t, we have

$$f(\theta^t) - f(\bar{\theta}) \leq \nabla f(\theta^t)^\top (\theta^t - \bar{\theta}). \tag{26}$$

Consider the SGD update $\theta^{t+1} = \theta^t - \alpha_t \nabla h(\theta^t, z_{i_t})$, where the sample z_{i_t} is drawn uniformly at random from the training set S, ensuring that the stochastic gradient is unbiased, i.e., $\mathbb{E}\left[\nabla h\left(\theta^t, z_{i_t}\right)\right] = \nabla f\left(\theta^t\right)$. Then,

$$\|\theta^{t+1} - \bar{\theta}\|^2 = \|\theta^t - \bar{\theta}\|^2 - 2\alpha_t(\theta^t - \bar{\theta})^\top \nabla h(\theta^t, z_{i_t}) + \alpha_t^2 \|\nabla h(\theta^t, z_{i_t})\|^2.$$

Taking expectation and using the unbiasedness of stochastic gradients, we obtain

$$\mathbb{E}\|\theta^{t+1} - \bar{\theta}\|^2 = \mathbb{E}\|\theta^t - \bar{\theta}\|^2 - 2\alpha_t \mathbb{E}[(\theta^t - \bar{\theta})^\top \nabla f(\theta^t)] + \alpha_t^2 \mathbb{E}\|\nabla h(\theta^t, z_{i_t})\|^2.$$

For fixed z_{i_t}, if $\tilde{\theta}$ is the optimizer of SGD, then $\nabla h(\tilde{\theta}, z_{i_t}) = 0$. As $h(\theta, z)$ is C_θ-Hölder continuous and η-approximately β-Hölder smooth, and $\|\theta^t - \tilde{\theta}\| \leq R$, it follows that

$$\begin{aligned} \mathbb{E}\left\|\nabla h(\theta^t, z_{i_t})\right\|^2 &= \mathbb{E}\left\|\nabla h(\theta^t, z_{i_t}) - \nabla h(\tilde{\theta}, z_{i_t})\right\|^2 \\ &\leq \mathbb{E}\left(L_\theta \|\theta^t - \tilde{\theta}\|^m + 2L_z \epsilon^{m'}\right)^2 \leq \left(L_\theta R^m + 2L_z \epsilon^{m'}\right)^2. \end{aligned} \tag{27}$$

Combining Eqs. (26) and (27), we obtain

$$\mathbb{E}\|\theta^{t+1} - \bar{\theta}\|^2 \leq \mathbb{E}\|\theta^t - \bar{\theta}\|^2 - 2\alpha_t \mathbb{E}[f(\theta^t) - f(\bar{\theta})] + \alpha_t^2 \left(L_\theta R^m + 2L_z \epsilon^{m'}\right)^2.$$

Rearranging terms yields

$$2\alpha_t \mathbb{E}\left[f(\theta^t) - f(\bar{\theta})\right] \leq \mathbb{E}\|\theta^t - \bar{\theta}\|^2 - \mathbb{E}\|\theta^{t+1} - \bar{\theta}\|^2 + \alpha_t^2 \left(L_\theta R^m + 2L_z \epsilon^{m'}\right)^2.$$

Summing the above inequality from $t = 1$ to T, we have

$$\begin{aligned}
\sum_{t=1}^{T} 2\alpha_t \mathbb{E}\left[f(\theta^t) - f(\bar{\theta})\right] &\leq \sum_{t=1}^{T} \left(\mathbb{E}\left[\|\theta^t - \bar{\theta}\|^2\right] - \mathbb{E}\left[\|\theta^{t+1} - \bar{\theta}\|^2\right]\right) + \left(L_\theta R^m + 2L_z \epsilon^{m'}\right)^2 \\
&\quad \sum_{t=1}^{T} \alpha_t^2 \\
&\leq \mathbb{E}\left[\|\theta^1 - \bar{\theta}\|^2\right] - \mathbb{E}\left[\|\theta^{T+1} - \bar{\theta}\|^2\right] + \left(L_\theta R^m + 2L_z \epsilon^{m'}\right)^2 \\
&\quad \sum_{t=1}^{T} \alpha_t^2 \\
&\leq D^2 + \left(L_\theta R^m + 2L_z \epsilon^{m'}\right)^2 \sum_{t=1}^{T} \alpha_t^2.
\end{aligned}$$

Therefore, there exists some $k \leq T$ such that

$$\mathcal{E}_{\text{opt}}(\theta^k) = \mathbb{E}[f(\theta^k) - f(\bar{\theta})] \leq \frac{D^2 + \left(L_\theta R^m + 2L_z \epsilon^{m'}\right)^2 \sum_{t=1}^{T} \alpha_t^2}{2\sum_{t=1}^{T} \alpha_t}.$$

This completes the proof. □

Theorem 3 reveals the relationship between the error of the optimization process and the step size α_t, initialization D, and the constant $L_\theta R^m + 2L_z \epsilon^{m'}$ in the adversarial loss with Hölder continuous and approximate Hölder smooth. In particular, if $\alpha_t = 1/\sqrt{T}$, then $\mathcal{E}_{opt} \leq \mathcal{O}(1/\sqrt{T})$, which is the convergence rate of SGD on the convex setting. A representative example satisfying Theorem 3 is the adversarial mean squared error (MSE) loss $h(\theta, z) = \max_{|\delta|_p \leq \epsilon} \left(\theta^T(x + \delta) - y\right)^2$ for linear regression models. For any fixed δ, the inner loss is convex with respect to θ, and hence its pointwise maximum over δ remains convex. Therefore, this adversarial MSE loss satisfies the convexity assumption in Theorem 3.

ii) **Non-convex adversarial loss**

Theorem 4 *(Optimization error under non-convex adversarial loss) Let $m, m' \in (0, 1]$, and $\Theta \subset \mathbb{R}^d$ be a bounded set with diameter at most R. Assume that $h(\theta, z)$ is C_θ-Hölder continuous, and η-approximately β-Hölder smooth on Θ for all $z \in \mathcal{Z}$. Let $\bar{\theta}$ be an optimal solution and $D = \|\theta^1 - \bar{\theta}\|$ as the distance between the initialization θ^1 and the optimal solution. Consider running SGD for T iterations with step sizes $\{\alpha_t\}_{t=1}^T$. Then, there exists an $k \leq T$ such that*

$$\mathcal{E}_{\text{opt}}(\theta^k) \leq \frac{D^2 + (L_\theta R^m + 2L_z \epsilon^{m'})^2 \sum_{t=1}^{T} \alpha_t^2}{2\sum_{t=1}^{T} \alpha_t} + M, \tag{28}$$

where $M = \beta R^{m+1} + \eta R$.

Proof Since $h(\theta, z)$ is C_θ-Hölder and η-approximately β-Hölder smooth with respect to θ for any fixed z, the empirical risk $R_S(\theta) = \frac{1}{n}\sum_{i=1}^{n} h(\theta, z_i)$ also satisfies both conditions. For convenience of notation, let $f(\theta) = R_S(\theta)$. Therefore, for any t, we have

$$\begin{aligned} f(\theta^t) - f(\bar{\theta}) &= \int_0^1 \langle \nabla f(\bar{\theta} + t(\theta^t - \bar{\theta})), \theta^t - \bar{\theta} \rangle dt \\ &= \langle \nabla f(\theta^t), \theta^t - \bar{\theta} \rangle + \int_0^1 \langle \nabla f(\bar{\theta} + t(\theta^t - \bar{\theta})) - \nabla f(\theta^t), \theta^t - \bar{\theta} \rangle dt. \end{aligned}$$

Since $f(\theta)$ is η-approximately β-Hölder smooth, it follows that

$$\|\nabla f(\bar{\theta} + t(\theta^t - \bar{\theta})) - \nabla f(\theta^t)\| \leq \beta \|(t-1)(\theta^t - \bar{\theta})\|^m + \eta.$$

Then we have

$$\begin{aligned} f(\theta^t) - f(\bar{\theta}) &\leq \langle \nabla f(\theta^t), \theta^t - \bar{\theta} \rangle + \int_0^1 \|\nabla f(\bar{\theta} + t(\theta^t - \bar{\theta})) - \nabla f(\theta^t)\| \cdot \|\theta^t - \bar{\theta}\| dt \\ &\leq \langle \nabla f(\theta^t), \theta^t - \bar{\theta} \rangle + \int_0^1 \left[\beta \big((t-1)\|\theta^t - \bar{\theta}\| \big)^m + \eta \right] \cdot \|\theta^t - \bar{\theta}\| dt \\ &= \nabla f(\theta^t)^T (\theta^t - \bar{\theta}) + \frac{\beta}{m+1} \|\theta^t - \bar{\theta}\|^{m+1} + \eta \|\theta^t - \bar{\theta}\| \\ &\leq \nabla f(\theta^t)^T (\theta^t - \bar{\theta}) + \frac{\beta}{m+1} R^{m+1} + \eta R. \end{aligned}$$

Reorganizing this expression yields

$$\nabla f(\theta^t)^T (\theta^t - \bar{\theta}) \geq f(\theta^t) - f(\bar{\theta}) - \frac{\beta}{m+1} R^{m+1} - \eta R. \tag{29}$$

Consider the SGD update $\theta^{t+1} = \theta^t - \alpha_t \nabla h(\theta^t, z_{i_t})$, where $\mathbb{E}[\nabla h(\theta^t, z_{i_t})] = \nabla f(\theta^t)$. Then,

$$\|\theta^{t+1} - \bar{\theta}\|^2 = \|\theta^t - \bar{\theta}\|^2 - 2\alpha_t (\theta^t - \bar{\theta})^\top \nabla h(\theta^t, z_{i_t}) + \alpha_t^2 \|\nabla h(\theta^t, z_{i_t})\|^2.$$

Taking expectation and using the unbiasedness of stochastic gradients, we obtain

$$\mathbb{E}\|\theta^{t+1} - \bar{\theta}\|^2 = \mathbb{E}\|\theta^t - \bar{\theta}\|^2 - 2\alpha_t \mathbb{E}[(\theta^t - \bar{\theta})^\top \nabla f(\theta^t)] + \alpha_t^2 \mathbb{E}\|\nabla h(\theta^t, z_{i_t})\|^2.$$

As $h(\theta, z)$ is C_θ-Hölder continuous and η-approximately β-Hölder smooth, and $\|\theta^t - \bar{\theta}\| \leq R$, similar to the proof process of Theorem 3, it follows that

$$\mathbb{E}\left\| \nabla h(\theta^t, z_{i_t}) \right\|^2 \leq (L_\theta R^m + 2L_z \epsilon^{m'})^2. \tag{30}$$

Combining Eqs. (29) and (30), we obtain

$$\mathbb{E}\|\theta^{t+1}-\bar{\theta}\|^2 \le \mathbb{E}\|\theta^t-\bar{\theta}\|^2 - 2\alpha_t\mathbb{E}[f(\theta^t)-f(\bar{\theta})-\frac{\beta}{m+1}R^{m+1}-\eta R] + \alpha_t^2(L_\theta R^m + 2L_z\epsilon^{m'})^2.$$

Rearranging terms yields

$$2\alpha_t\mathbb{E}[f(\theta^t)-f(\bar{\theta})] \le \mathbb{E}\|\theta^t-\bar{\theta}\|^2 - \mathbb{E}\|\theta^{t+1}-\bar{\theta}\|^2 + \alpha_t(\frac{2\beta}{m+1}R^{m+1}+2\eta R) + \alpha_t^2(L_\theta R^m + 2L_z\epsilon^{m'})^2.$$

Summing the above inequality from $t=1$ to T, we have

$$\begin{aligned}
&\sum_{t=1}^{T} 2\alpha_t\mathbb{E}[f(\theta^t)-f(\bar{\theta})] \\
\le&\sum_{t=1}^{T}\left(\mathbb{E}[\left\|\theta^t-\bar{\theta}\right\|^2]-\mathbb{E}[\left\|\theta^{t+1}-\bar{\theta}\right\|^2]\right)+\sum_{t=1}^{T}\alpha_t(\frac{2\beta}{m+1}R^{m+1}+2\eta R) \\
&+(L_\theta R^m+2L_z\epsilon^{m'})^2\sum_{t=1}^{T}\alpha_t^2 \\
\le&\mathbb{E}[\left\|\theta^1-\bar{\theta}\right\|^2]-\mathbb{E}[\left\|\theta^{T+1}-\bar{\theta}\right\|^2]+\sum_{t=1}^{T}\alpha_t(\frac{2\beta}{m+1}R^{m+1}+2\eta R) \\
&+(L_\theta R^m+2L_z\epsilon^{m'})^2\sum_{t=1}^{T}\alpha_t^2 \\
\le&D^2+\sum_{t=1}^{T}\alpha_t(\frac{2\beta}{m+1}R^{m+1}+2\eta R)+(L_\theta R^m+2L_z\epsilon^{m'})^2\sum_{t=1}^{T}\alpha_t^2.
\end{aligned}$$

Therefore, there exists some $k \le T$ such that

$$\begin{aligned}
\mathcal{E}_{\text{opt}}(\theta^k)=\mathbb{E}[f(\theta^k)-f(\bar{\theta})] &\le \frac{D^2+(L_\theta R^m+2L_z\epsilon^{m'})^2\sum_{t=1}^{T}\alpha_t^2}{2\sum_{t=1}^{T}\alpha_t}+\frac{\frac{2\beta}{m+1}R^{m+1}+2\eta R}{2} \\
&\le \frac{D^2+(L_\theta R^m+2L_z\epsilon^{m'})^2\sum_{t=1}^{T}\alpha_t^2}{2\sum_{t=1}^{T}\alpha_t}+M,
\end{aligned}$$

where $M=\beta R^{m+1}+\eta R$. This completes the proof. □

Theorem 4 extends the analysis to non-convex adversarial training. It characterizes the relationship between the optimization error and the step size α_t, initialization D, and other constants. Unlike the convex case, the resulting upper bound contains an additional constant term $M=\beta R^{m+1}+\eta R$, where $\beta=L_\theta$ and $\eta=2L_z\epsilon^{m'}$ according to Lemma 4(ii). Consequently, when running SGD on a non-convex adversarial loss, the optimization error cannot

converge exactly to zero but instead is ultimately bounded by a residual term determined by M. In particular, choosing the step size $\alpha_t = 1/\sqrt{T}$ yields $\mathcal{E}_{\text{opt}} \leq \mathcal{O}(1/\sqrt{T}) + M$.

The residual term $M = L_\theta R^{m+1} + 2L_z\epsilon^{m'}R$ is explicitly determined by the Hölder smoothness constant L_θ, the parameter-space diameter R, and the adversarial perturbation radius ϵ. It reflects the price paid for relaxing the exact Hölder smoothness assumption and quantifies the attainable optimization accuracy under the approximate Hölder smoothness assumption for adversarial training. Since the parameter space Θ is assumed to be bounded and the adversarial perturbation radius ϵ is typically chosen to be small in practice, M remains explicitly controlled by these problem-dependent quantities. Consequently, a smaller perturbation radius or a tighter parameter domain leads to a smaller residual term and a tighter optimization bound. Moreover, an example satisfying Theorem 4 is the adversarial cross-entropy loss with q-norm regularization term ($1 \leq q \leq 2$), formulated as $h(\theta, z) = \max_{\|\delta\|_p \leq \epsilon} \mathcal{L}_{CE}(f_\theta(x+\delta), y) + \lambda\|\theta\|^q$. Since f_θ is highly nonlinear with respect to the model parameters, the resulting adversarial objective is generally non-convex in θ. For standard losses in neural networks, such as the cross-entropy loss, the Lipschitz continuity of the gradient typically holds under mild boundedness assumptions on the model parameters (Gao et al., 2022). Meanwhile, the q-norm regularization term ($1 < q \leq 2$) satisfies the $(q-1, L)$-Hölder smoothness condition. Therefore, this objective $h(\theta, z)$ provides a non-convex example in Theorem 4.

4.3 Excess Risk Minimization

We have now established bounds for both generalization and optimization errors in adversarial training. According to the excess risk decomposition in Eq. (2), the excess risk satisfies

$$\text{Excess Risk} \leq \mathcal{E}_{\text{gen}} + \mathcal{E}_{\text{opt}}.$$

Thus, improving model performance requires minimizing the sum of generalization and optimization errors. To achieve this, it is necessary to balance the effects of the step size α_t and the number of iterations T. However, optimizing both simultaneously is challenging. Here, we only consider the simple case where the step size is constant $\alpha_t = \alpha$ for convex and non-convex losses.

i) **Excess risk on convex adversarial loss**. Combining Eqs. (17) and (25), we obtain

$$\begin{aligned}
&\mathcal{E}_{\text{gen}} + \mathcal{E}_{\text{opt}} \lesssim \\
&C_\theta\Big[(\sqrt{2}\eta + \sqrt{\frac{2C(1-m)}{1+m}})T\alpha + \frac{(L_\theta R^m + 2L_z\epsilon^{m'})T\alpha}{n}\Big]^m + \frac{D^2}{2T\alpha} \\
&\quad + \frac{(L_\theta R^m + 2L_z\epsilon^{m'})^2\alpha}{2}.
\end{aligned} \tag{31}$$

In Eq. (31), the second term is controlled by T, while the last term is essentially fixed for small α. The first term has two components: one controlled by the sample size n, which becomes small for a large training set, and another involving $(\sqrt{2}\eta + \sqrt{\frac{2C(1-m)}{1+m}})T\alpha$, which leads to robust overfitting when T grows large. This suggests that the $\eta = 2L_z\epsilon^{m'}$ approximate Hölder smoothness of the adversarial loss explains robust overfitting. Therefore, early stopping is necessary for optimal performance. Empirically, early stopping has

been shown to be an effective technique for adversarial training (Grabinski et al., 2022). From Eq. (31), the optimal number of iterations T^* that minimizes the right-hand side is given by

$$T^* = \left[\frac{n^m \|\theta^1 - \theta^*\|^2}{2\, mC_\theta \alpha^{m+1} \left(n(\sqrt{2}\eta + \sqrt{\frac{2C(1-m)}{1+m}}) + L_\theta R^m + 2L_z \epsilon^{m'} \right)^m} \right]^{\frac{1}{m+1}}.$$

Since T^* is unknown in practice, selecting stopping criteria is essential. A validation set can be used to determine the optimal stopping point and prevent robust overfitting.

ii) **Excess risk on non-convex adversarial loss**. For non-convex adversarial loss, we also consider the fixed step size $\alpha_t = \alpha$. Combining Eqs. (22) and (28), we have

$$\mathcal{E}_{\text{gen}} + \mathcal{E}_{\text{opt}} \lesssim \frac{B}{n} + C_\theta \left(\eta + \beta R^m + \frac{L_\theta R^m + 2L_z \epsilon^{m'}}{n} \right)^m (T\alpha)^m + \frac{D^2}{2T\alpha} + \frac{(L_\theta R^m + 2L_z \epsilon^{m'})^2 \alpha}{2} + M. \tag{32}$$

We can similarly analyze Eq. (32) to show that the second term may lead to robust overfitting. By minimizing the right-hand side of the bound, we can also derive the optimal number of iterations T^*, which is given by

$$T^* = \left[\frac{n^m \|\theta^1 - \theta^*\|^2}{2\, mC_\theta \alpha^{m+1} \left(n(\eta + \beta R^m) + L_\theta R^m + 2L_z \epsilon^{m'} \right)^m} \right]^{\frac{1}{m+1}}.$$

Similarly, T^* is also difficult to compute in practice and a validation set is usually used to determine the optimal stopping point to prevent robust overfitting.

5 Example

This section considers a practical example based on Hölder-type conditions and presents a series of experiments to validate the theoretical results established in the preceding sections. Our primary objective is to examine how the Hölder-type conditions influence the robust generalization behavior of adversarial training, which is quantified via the generalization gap. A smaller gap indicates stronger generalization ability. Accordingly, our goal is not to achieve state-of-the-art performance, but to clarify the specific impact of the Hölder-type conditions.

5.1 q-Norm Regularization: A Practical Hölder-Type Case

To connect our theoretical analysis with practical adversarial training, we consider a commonly used regularization strategy that naturally satisfies the Hölder-type conditions. We augment the loss function with a q-norm regularization term $R(\theta) = \|\theta\|^q$, whose gradient is m-Hölder continuous ($m = q - 1$) but is not Lipschitz continuous on Θ for $q \in [1, 2]$. Addi-

tionally, for standard losses in DNNs, such as the cross-entropy loss, Lipschitz continuity of the gradient typically holds under mild boundedness assumptions on the model parameters (Gao et al., 2022). Therefore, when the cross-entropy loss is combined with $R(\theta)$, the resulting adversarial loss admits Hölder continuous and approximate Hölder smooth, i.e.,

$$\min h(\theta, z) \Leftrightarrow \min \max_{\|\delta\|_p \leq \epsilon} \mathcal{L}(\theta, x + \delta, y) + \lambda R(\theta), \tag{33}$$

where $\lambda > 0$, $\mathcal{L}(\theta, x, y)$ is the cross-entropy loss, and $z = (x, y)$ is the input–output pair.

This construction provides a concrete example that satisfies our theoretical assumptions and serves as the basis for the experimental evaluation presented in the following subsections.

5.2 Setup and Results

i) **Setup**. We adopt the PreActResNet-18 architecture for adversarial training, following the setup in (Rice et al., 2020; Xiao et al., 2022). Experiments are conducted on three image classification datasets: CIFAR-10[1] and CIFAR $-$ 100[1], and SVHN.[2] For the inner maximization problem, we employ the ℓ_∞-PGD adversarial training, following the approach of (Madry et al., 2018). The step size of the inner maximization is set to 0.5 for CIFAR-10 and 0.25 for CIFAR-100 and SVHN. The weight decay is fixed at 5×10^{-5}. The training loss is defined in Eq. (33), where $\lambda = 5 \times 10^{-5}$.

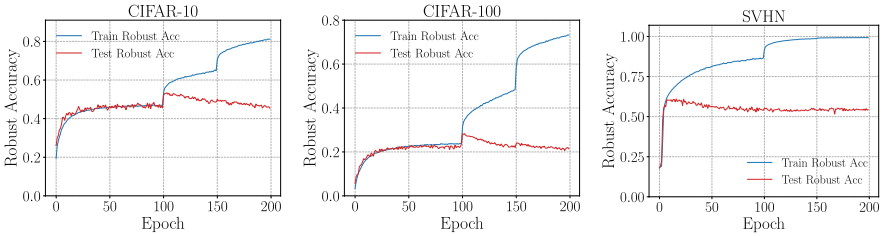
ii) **Robust overfitting phenomenon.** We train the models for 200 epochs under adversarial training with a fixed perturbation budget of $\epsilon = 8/255$ using the cross-entropy loss. On CIFAR-10 and CIFAR-100, the learning rate followed a piece-wise schedule of 0.1 for the first 100 epochs, 0.01 for the next 50 epochs, and 0.001 for the final 50 epochs. On SVHN, the learning rate started at 0.01 and followed the same piece-wise decay schedule. Figure 2 illustrates the evolution of robust training and test accuracies on three datasets. A clear and consistent robust overfitting phenomenon can be observed across all three datasets. Specifically, during the early stages of training, both robust training and test accuracies improve steadily, indicating that the models successfully learn adversarially robust features. However, after a certain point, typically around the first learning rate decay, the trajectories of training and test robust accuracy begin to diverge. While the training robust accuracy continues to increase, the test robust accuracy stagnates or even degrades, signaling the onset of robust overfitting. These results reveal that improvements in training performance do not necessarily translate into better test performance in adversarial training. This phenomenon highlights the heightened susceptibility of adversarially trained models to overfitting under perturbations, consistent with prior observations (Rice et al., 2020). Consequently, these findings underscore the necessity of effective regularization strategies to mitigate robust overfitting and improve robust generalization.

iii) **Robust generalization under Hölder-type loss**.

Effect of the exponent q ($m = q - 1$). We first study how the exponent q influences the robust generalization behavior. We empirically vary $q \in \{1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$ and

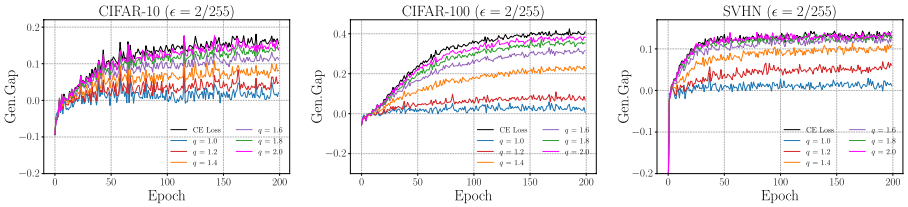
[1] https://www.cs.toronto.edu/~kriz/cifar.html

[2] http://ufldl.stanford.edu/housenumbers/



(a) CIFAR-10 (b) CIFAR-100 (c) SVHN

Fig. 2 Robust overfitting phenomenon on three datasets

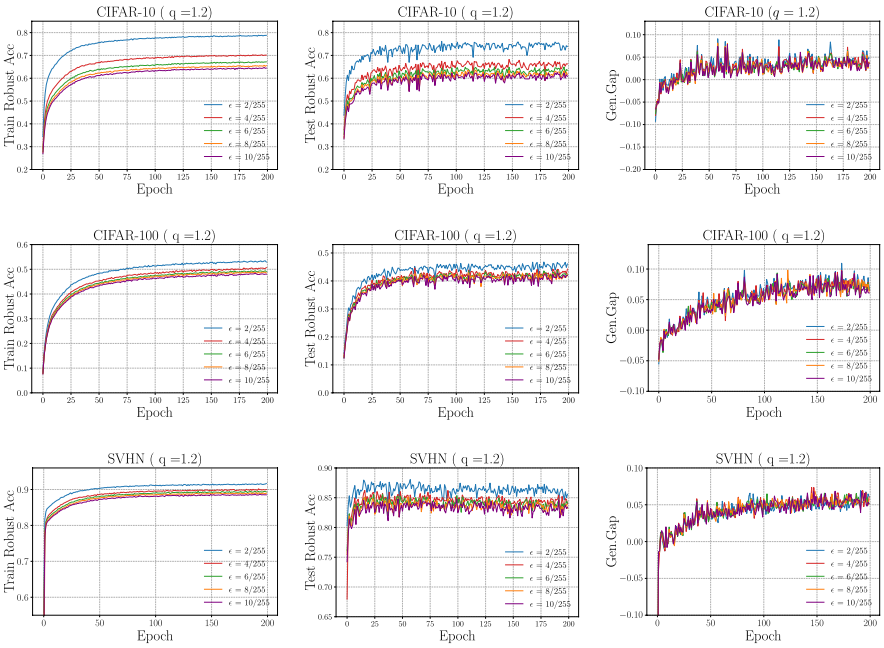


(a) CIFAR-10 (b) CIFAR-100 (c) SVHN

Fig. 3 Generalization gap with $q \in \{1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$ on three datasets, where $m = q - 1$. The black curve represents adversarial training using the cross-entropy loss

evaluate the generalization gap on CIFAR-10, CIFAR-100, and SVHN under a fixed perturbation radius $\epsilon = 2/255$ as an example. The results are shown in Fig. 3. It can be seen that the generalization gap exhibits a clear and consistent dependence on the exponent q across all three datasets. Compared with the baseline using the standard cross-entropy loss (shown as the black curve) for adversarial training, the Hölder-type loss consistently yields a smaller generalization gap throughout the training process. This observation indicates that introducing the q-norm regularization term effectively mitigates robust overfitting. More importantly, the generalization gap decreases monotonically as q is reduced from 2.0 to 1.0. This monotonic trend is particularly evident in the later stages of training, where robust overfitting typically becomes more pronounced. These empirical results are well aligned with our theoretical analysis, which predicts a tighter generalization bound for smaller values of q. Consequently, Fig. 3 provides strong empirical validation for the theoretical advantage of Hölder-type conditions in improving robust generalization.

Effect of perturbation radius ϵ . Next, we investigate the impact of the perturbation radius ϵ on robust training and test accuracies, and the generalization gap when empirically fixing $q = 1.2$, which offers a favorable trade-off between robustness and optimization stability. We vary $\epsilon \in \{2/255, 4/255, 6/255, 8/255, 10/255\}$ and report the results on CIFAR-10, CIFAR-100, and SVHN in Fig. 4. We observe a consistent trend across all datasets: as the perturbation radius increases, both training and test robust accuracy decrease. This behavior is expected, as larger perturbations correspond to stronger adversarial trends and thus result in a more challenging optimization problem. On the other hand, we observe that, under the fixed Hölder-type loss with $q = 1.2$, the generalization gap in Fig. 4 remains consistently small across different perturbation radii. Although slight fluctuations can be observed as ϵ



(a) Train robust accuracy (b) Test robust accuracy (c) Generalization Gap

Fig. 4 Performance comparison on CIFAR-10/CIFAR-100/SVHN with the q-norm regularization term $q = 1.2$ ($m = 0.2$) under perturbation radius $\epsilon \in \{2/255, 4/255, 6/255, 8/255, 10/255\}$

increases, no significant amplification of the gap occurs, even for relatively large perturbations. This result suggests that the Hölder-type loss induces stable training and maintains robust generalization over a wide range of adversarial perturbations. Rather than solely improving training robustness, the Hölder-type loss achieves a favorable balance between robustness and generalization.

6 Conclusion

This paper studies robust generalization in adversarial training from a theoretical perspective under general weak smoothness assumptions. We introduce Hölder continuity for adversarial losses and propose approximate Hölder smoothness as a new notion of weak smoothness. The stability of adversarial training is then analyzed. Under these conditions, we establish rigorous generalization and optimization gap bounds for SGD in both convex and non-convex settings, and further analyze the induced excess risk. Our results cover a broad class of losses in DNNs that do not satisfy global Lipschitz conditions but exhibit Hölder properties, and reveal the intrinsic connection between optimization and generalization in adversarial training, offering a theoretical perspective on robust overfitting. We also provide empirical validation through a concrete example, showing consistency between theory and practice. Future work will explore the Hölder-type condition analysis for advanced

optimization algorithms beyond SGD, and adaptive regularization schemes that dynamically adjust Hölder exponents during training.

Acknowledgements This work was supported by the National Natural Science Foundation of China under grants 62536006 and 12671639, and the Zhejiang Provincial Natural Science Foundation of China under grant LZ26F030003.

Author Contributions Hailiang Ye: Methodology, Formal analysis, Writing-original draft, Writing - review & editing, Funding acquisition; Ziqi An: Writing-original draft, Investigation, Software, Visualization, Data curation; Feilong Cao: Conceptualization, Writing - review & editing, Validation, Funding acquisition.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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